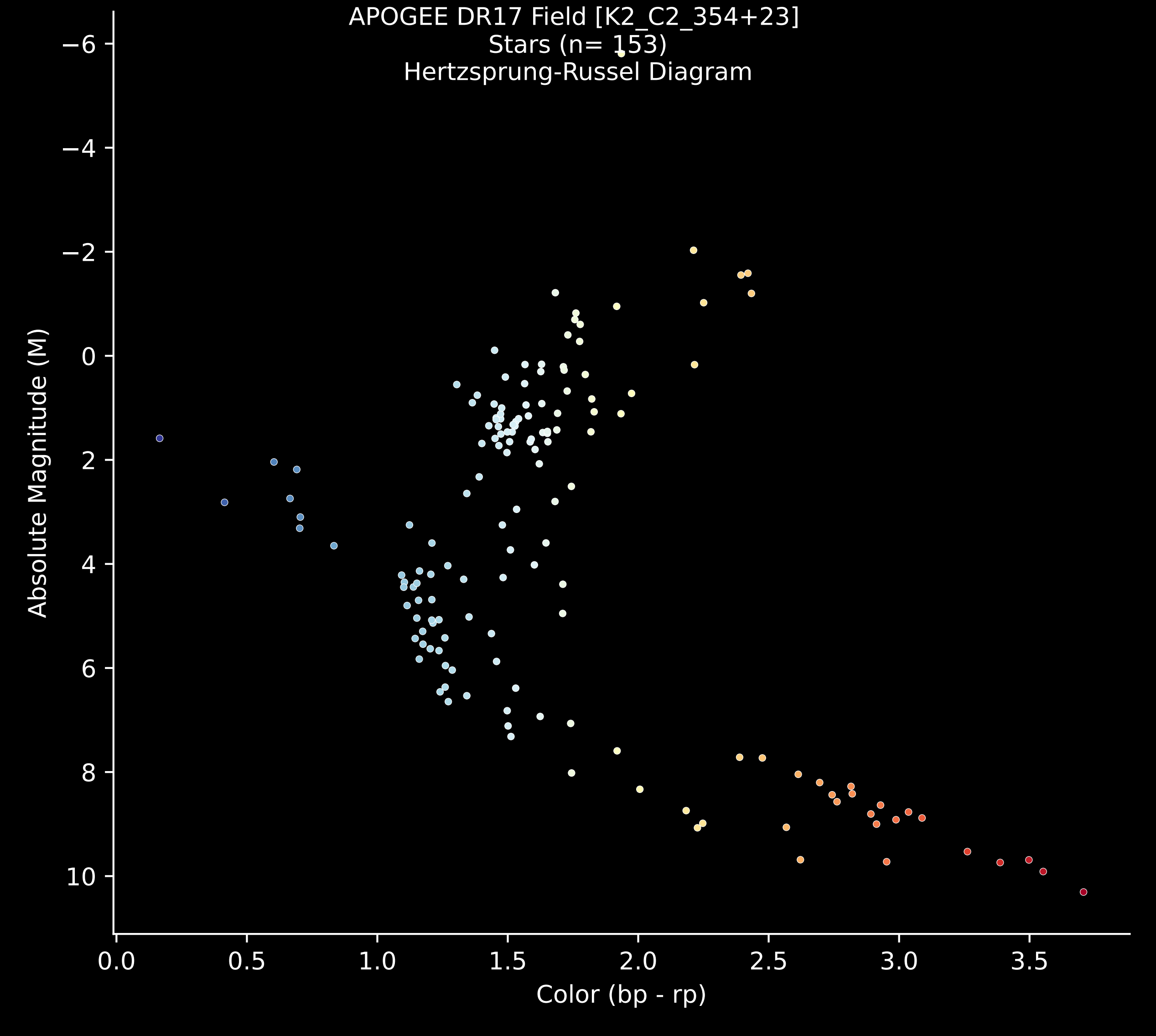
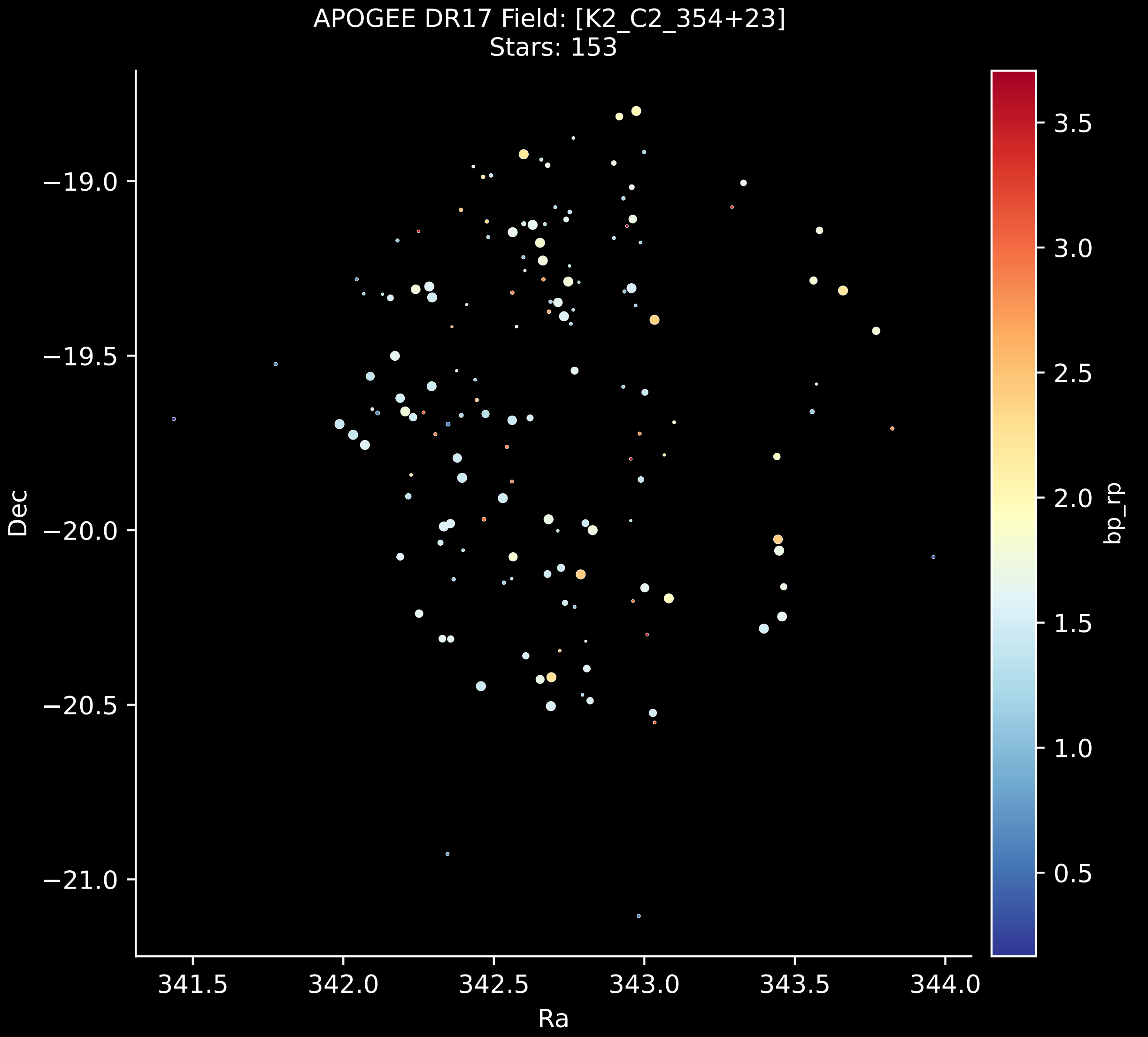
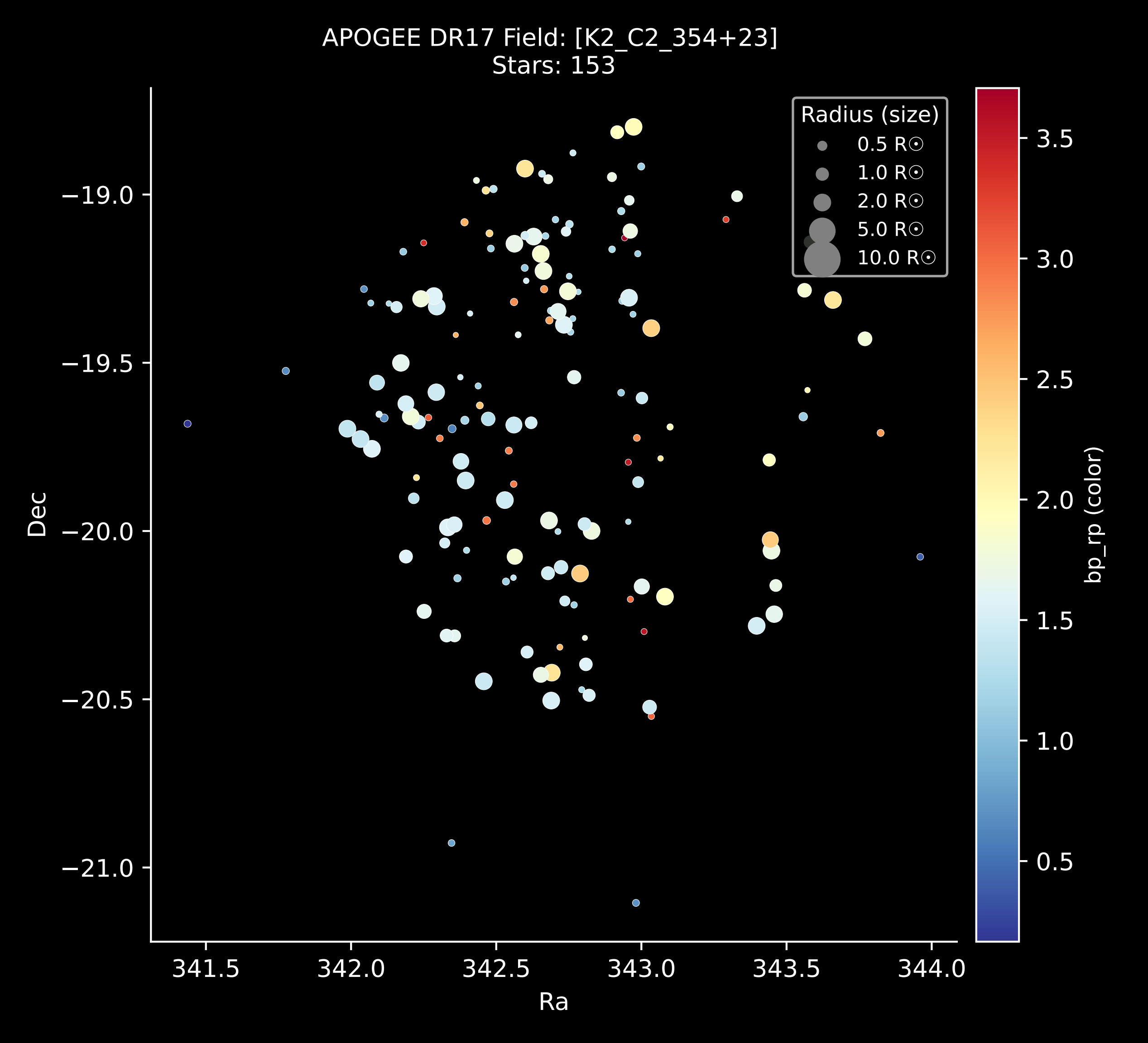
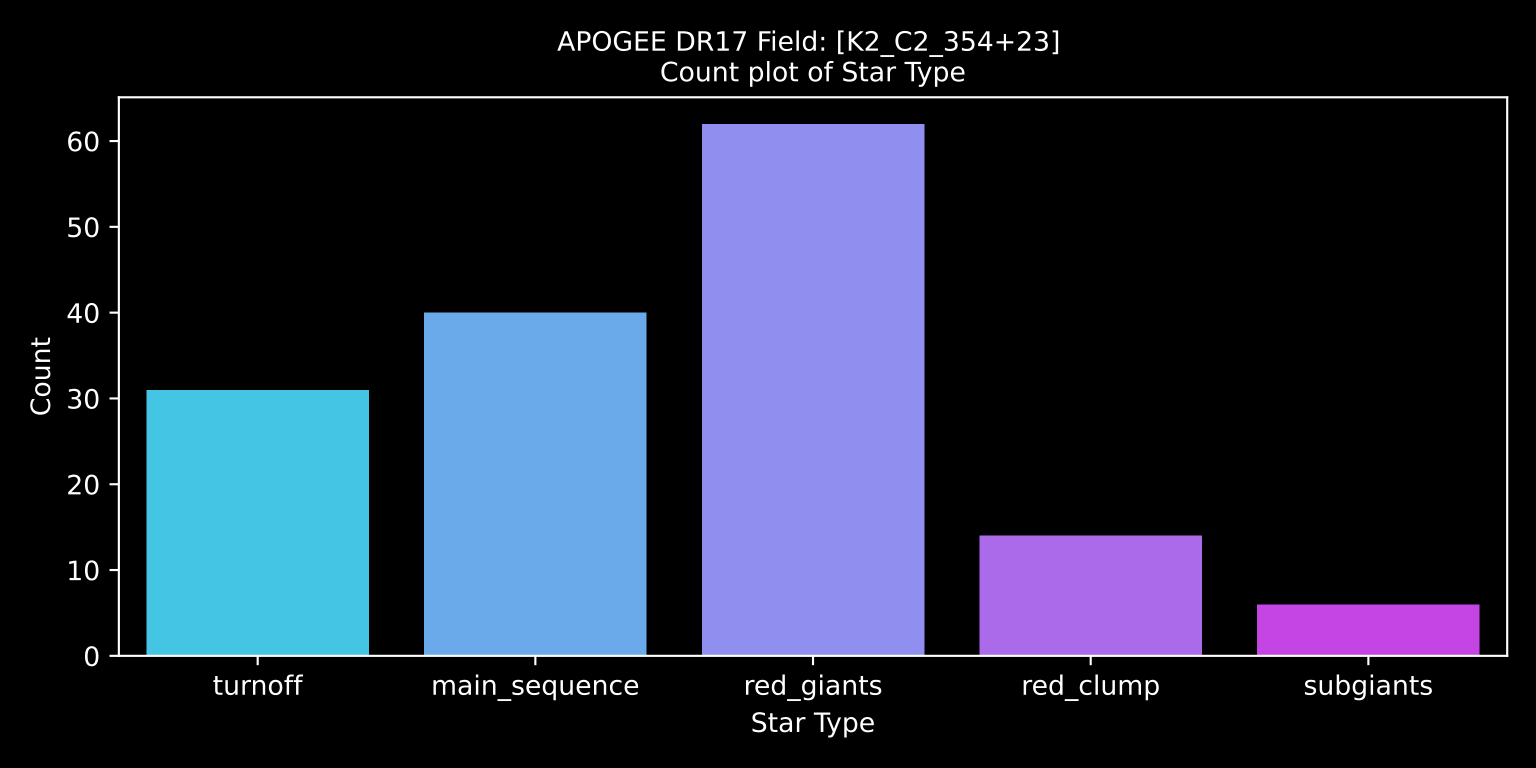
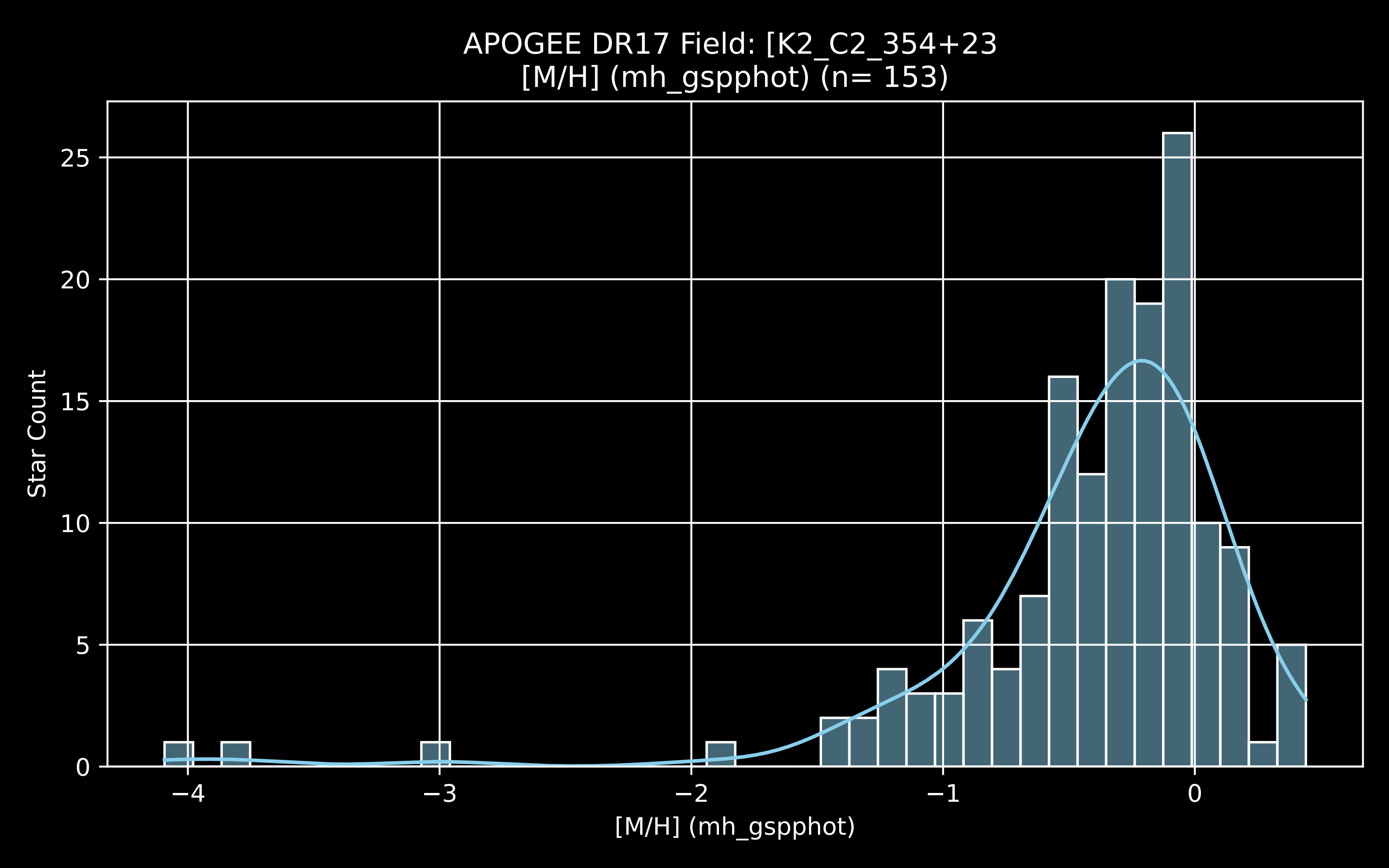


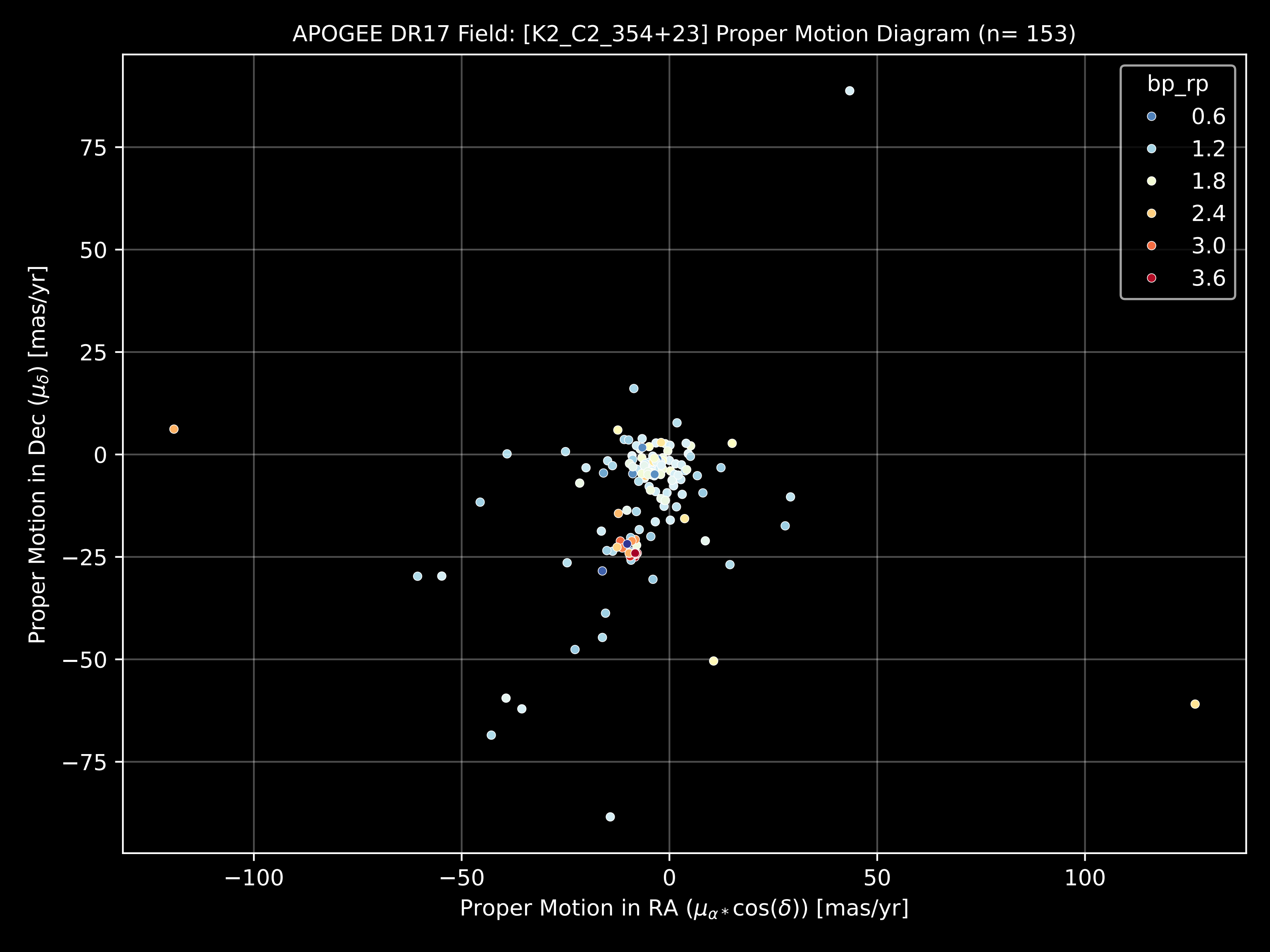
APOGEE DR17 Field [K2_C2_354+23]
Stars (n= 153)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−6
−4
−2
0
2
4
6
8
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5


APOGEE DR17 Field: [K2_C2_354+23]
Stars: 153
Dec
Ra
bp_rp
−19.0
−19.5
−20.0
−20.5
−21.0
341.5
342.0
342.5
343.0
343.5
344.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5

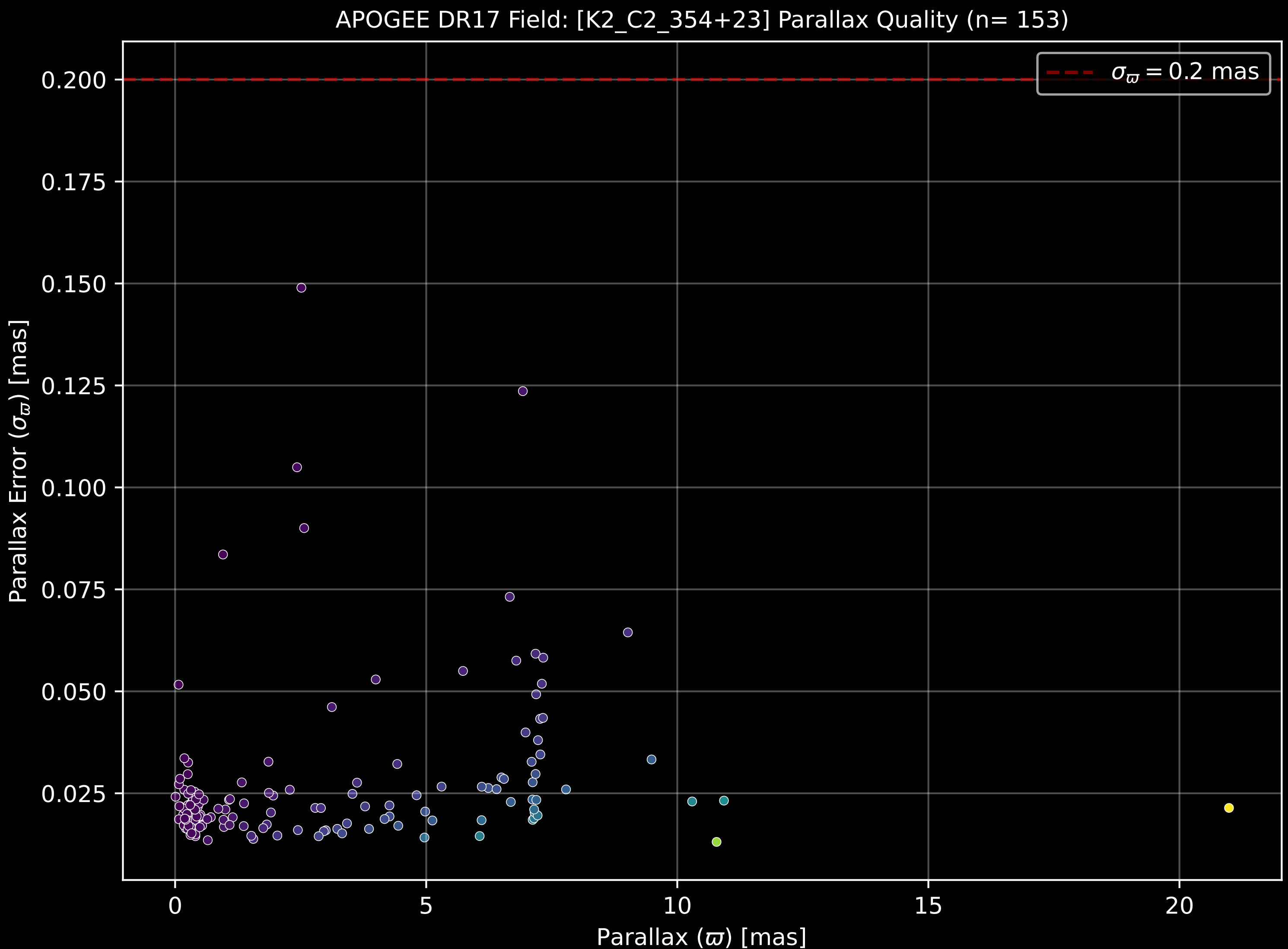

APOGEE DR17 Field: [K2_C2_354+23]
Stars: 153
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
−19.0
−19.5
−20.0
−20.5
−21.0
341.5
342.0
342.5
343.0
343.5
344.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5


APOGEE DR17 Field: [K2_C2_354+23]
Count plot of Star Type
Count
0
10
20
30
40
50
60
turnoff
main_sequence
red_giants
red_clump
subgiants
Star Type

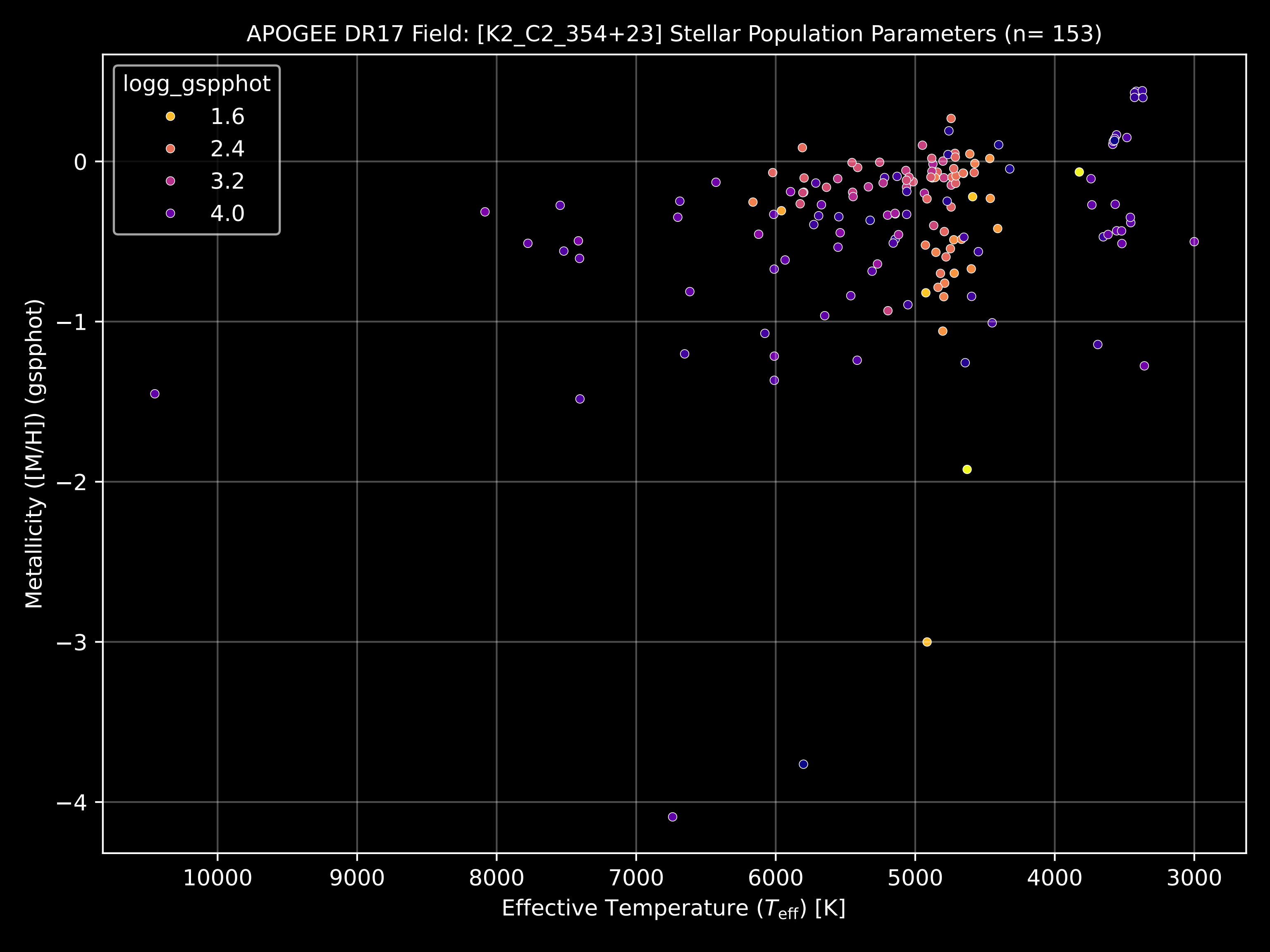

APOGEE DR17 Field: [K2_C2_354+23
[M/H] (mh_gspphot) (n= 153)
Star Count
25
20
15
10
5
0
−4
−3
−2
−1
0
[M/H] (mh_gspphot)



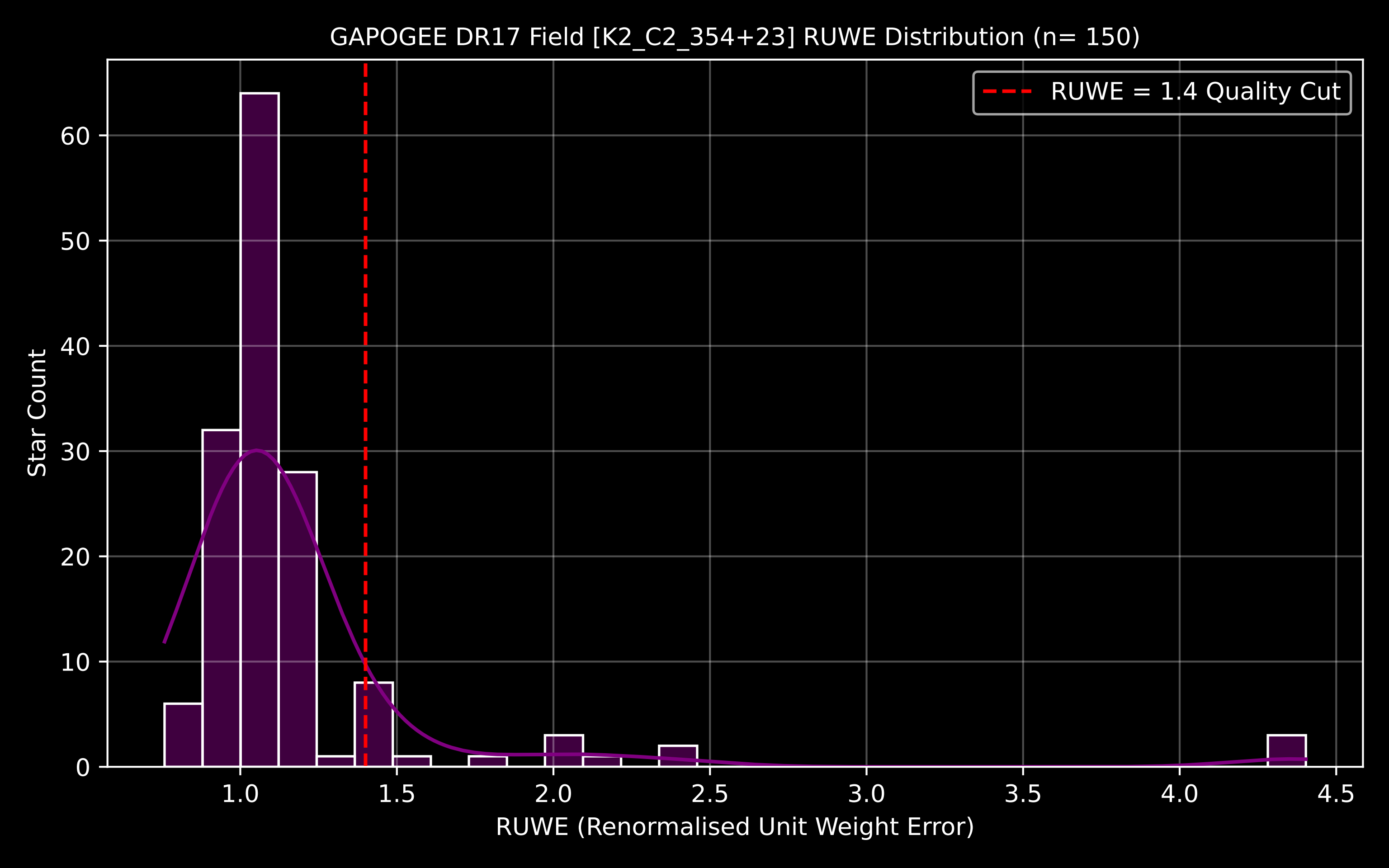
APOGEE DR17 Field: [K2_C2_354+23] Proper Motion Diagram (n= 153)
bp_rp
0.6
1.2
1.8
2.4
3.0
3.6
75
50
25
0
−25
−50
−75
−100
−50
0
50
100
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

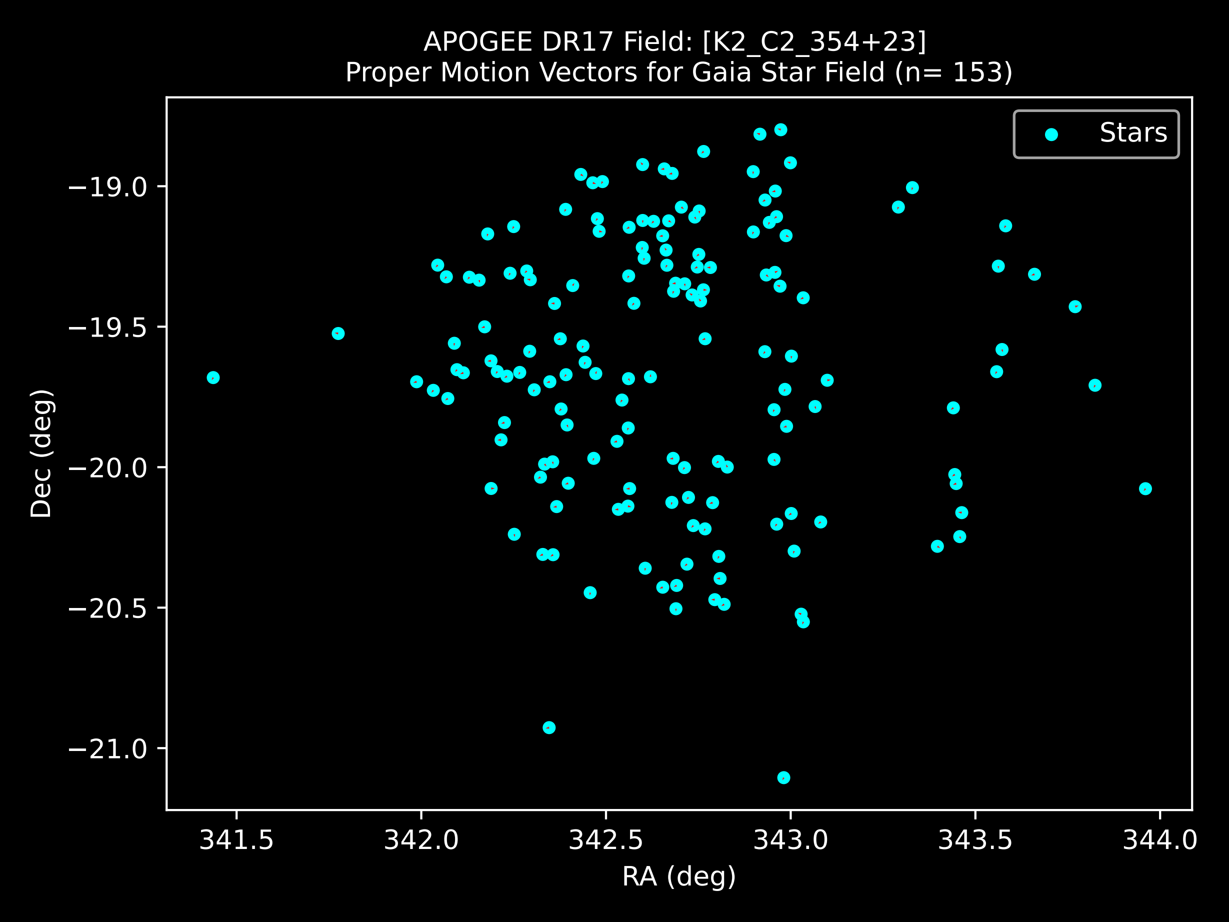


APOGEE DR17 Field: [K2_C2_354+23] Parallax Quality (n= 153)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0
5
10
15
20

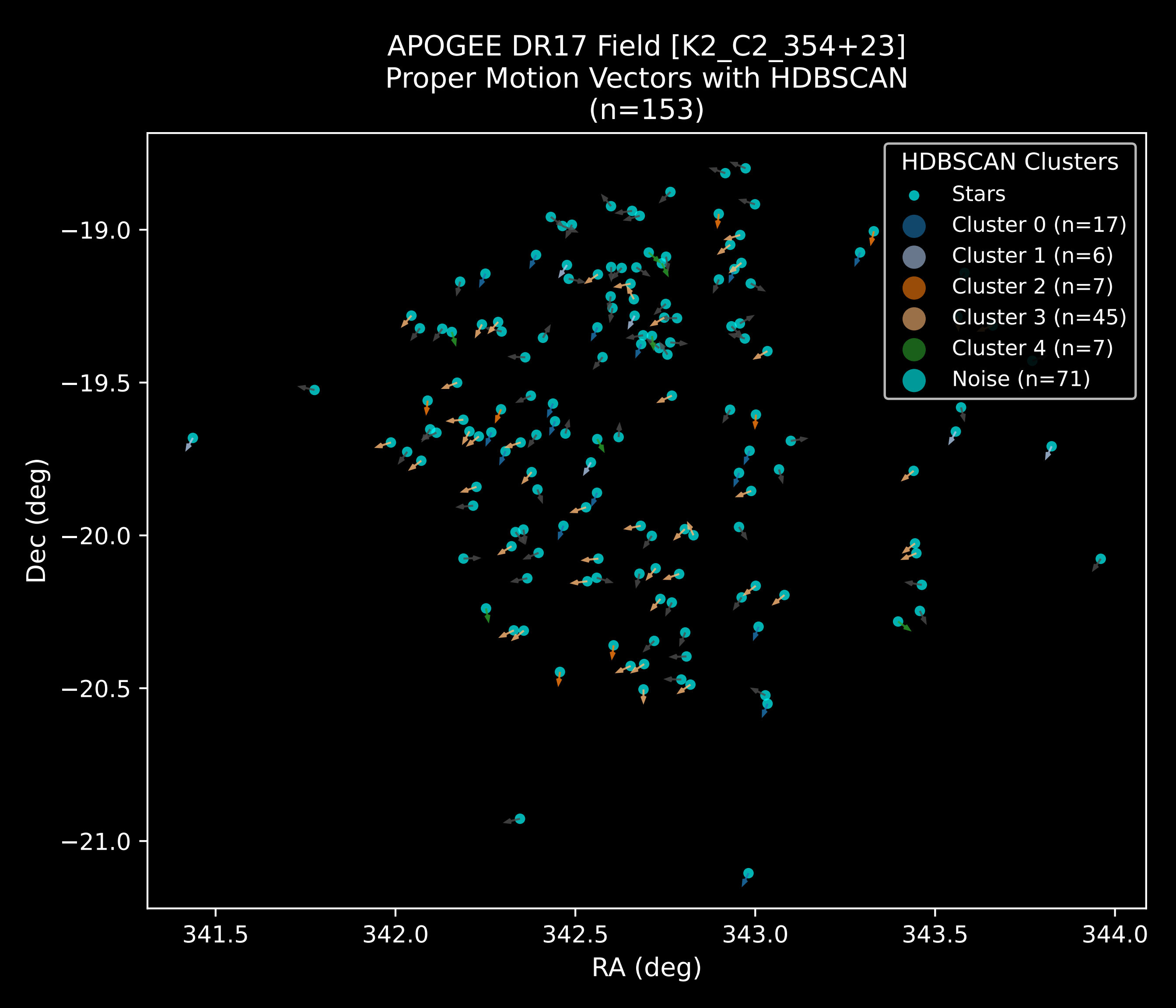


APOGEE DR17 Field: [K2_C2_354+23] Stellar Population Parameters (n= 153)
logg_gspphot
1.6
2.4
3.2
4.0
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
0
−1
−2
−3
−4
10000
9000
8000
7000
6000
5000
4000
3000

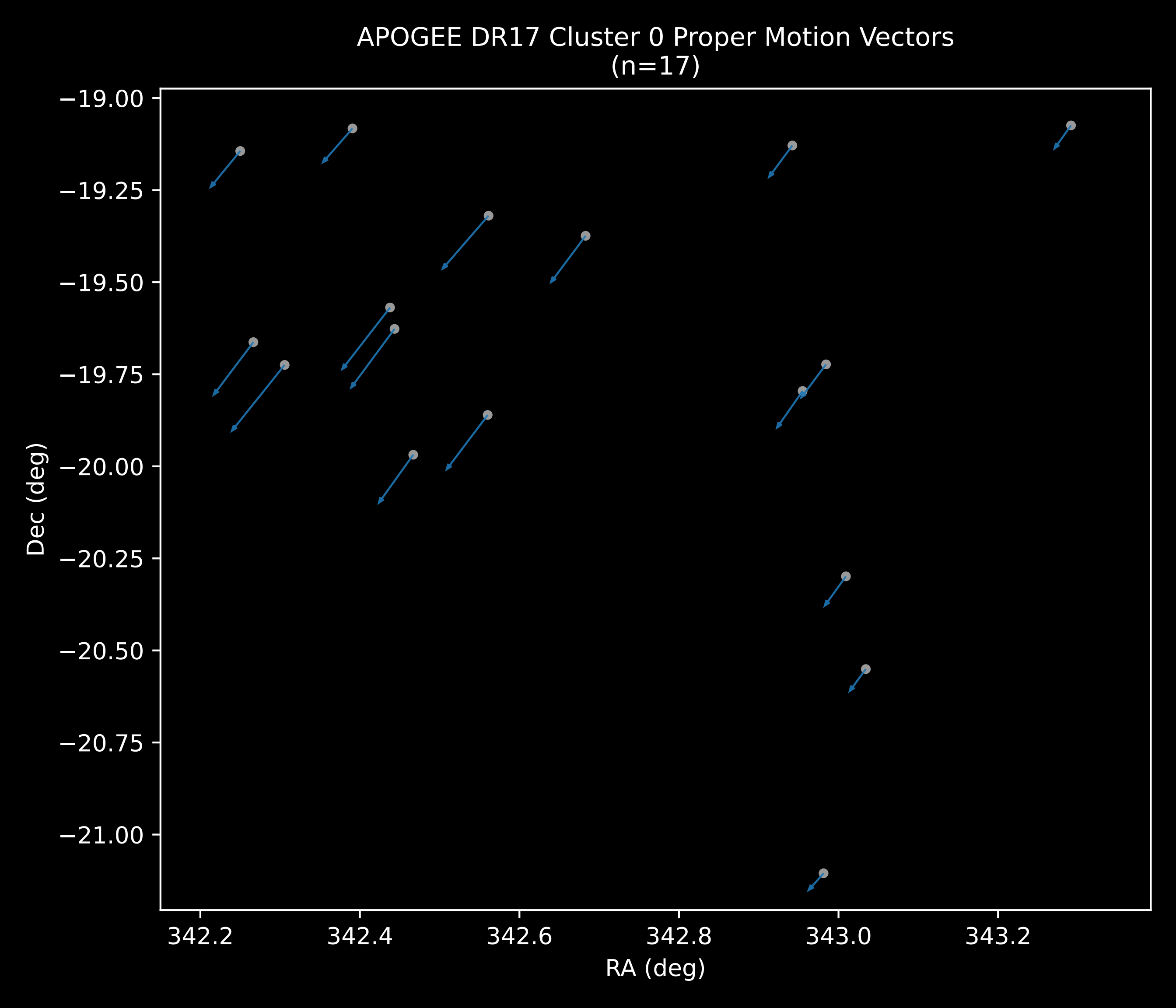

GAPOGEE DR17 Field [K2_C2_354+23] RUWE Distribution (n= 150)
RUWE = 1.4 Quality Cut
Star Count
0
10
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
RUWE (Renormalised Unit Weight Error)



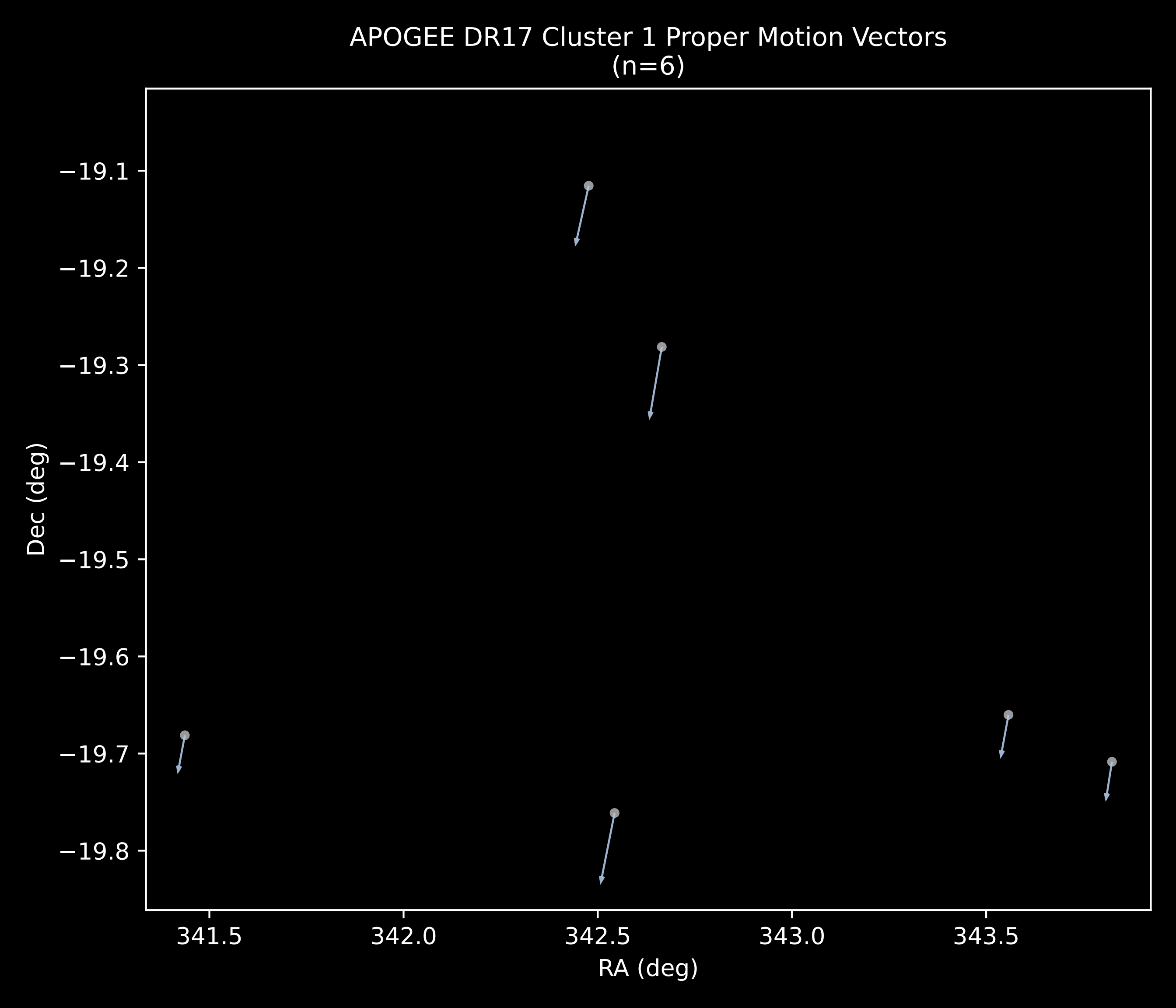
APOGEE DR17 Field: [K2_C2_354+23]
Proper Motion Vectors for Gaia Star Field (n= 153)
Stars
−19.0
−19.5
−20.0
−20.5
−21.0
Dec (deg)
341.5
342.0
342.5
343.0
343.5
344.0
RA (deg)



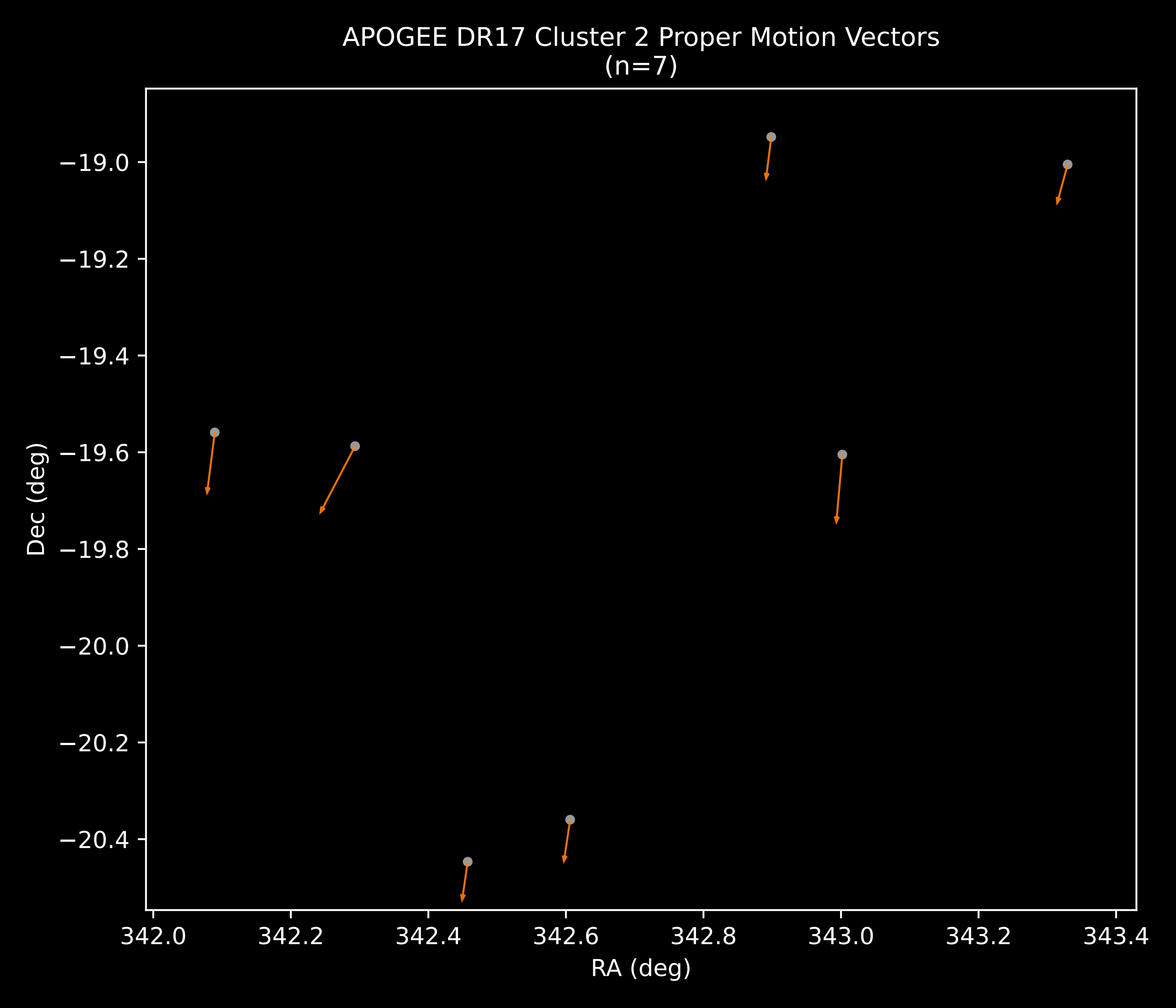
APOGEE DR17 Field [K2_C2_354+23]
Proper Motion Vectors with HDBSCAN
(n=153)
HDBSCAN Clusters
Stars
Cluster 0 (n=17)
Cluster 1 (n=6)
Cluster 2 (n=7)
Cluster 3 (n=45)
Cluster 4 (n=7)
Noise (n=71)
Dec (deg)
−19.0
−19.5
−20.0
−20.5
−21.0
RA (deg)
341.5
342.0
342.5
343.0
343.5
344.0

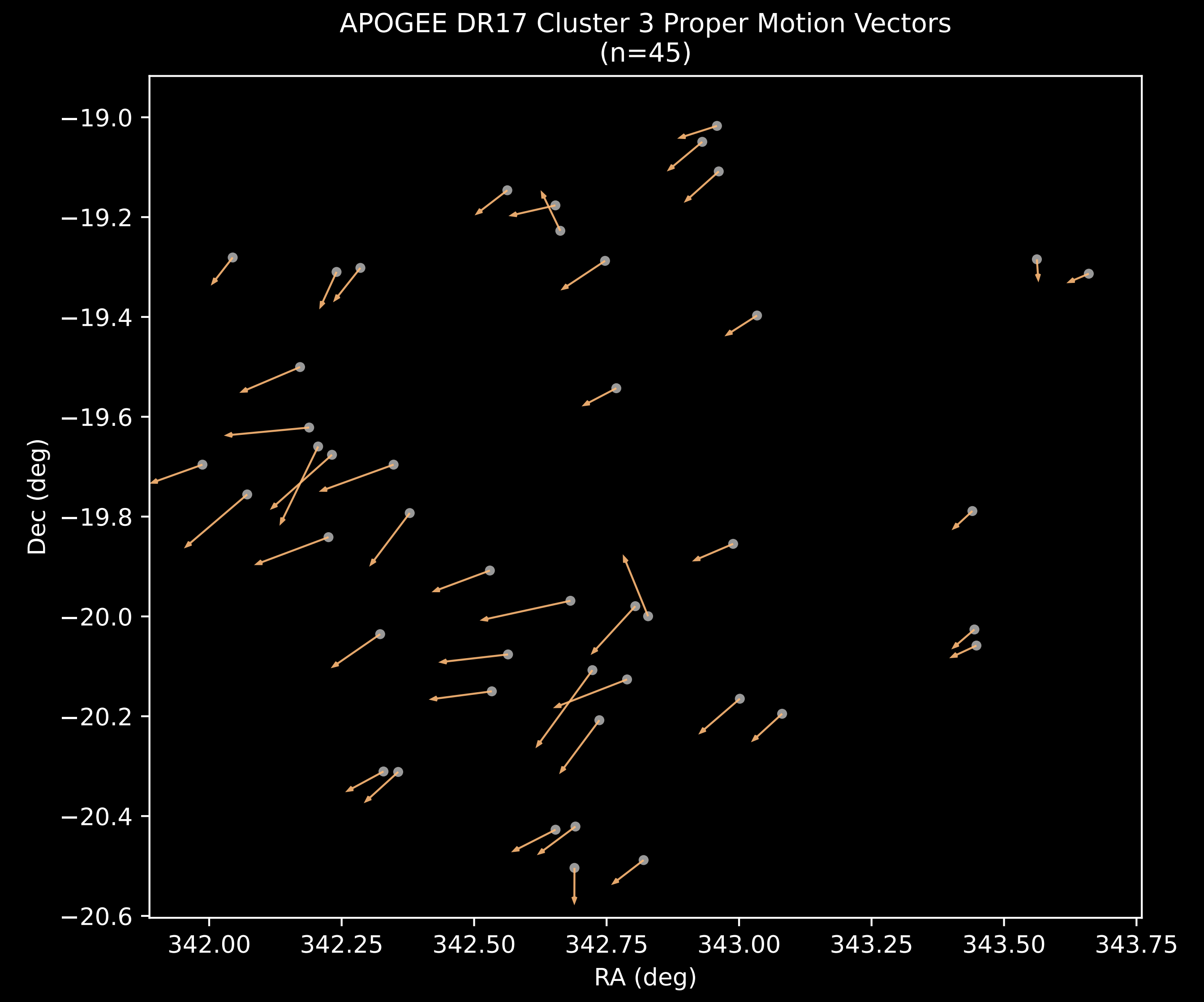


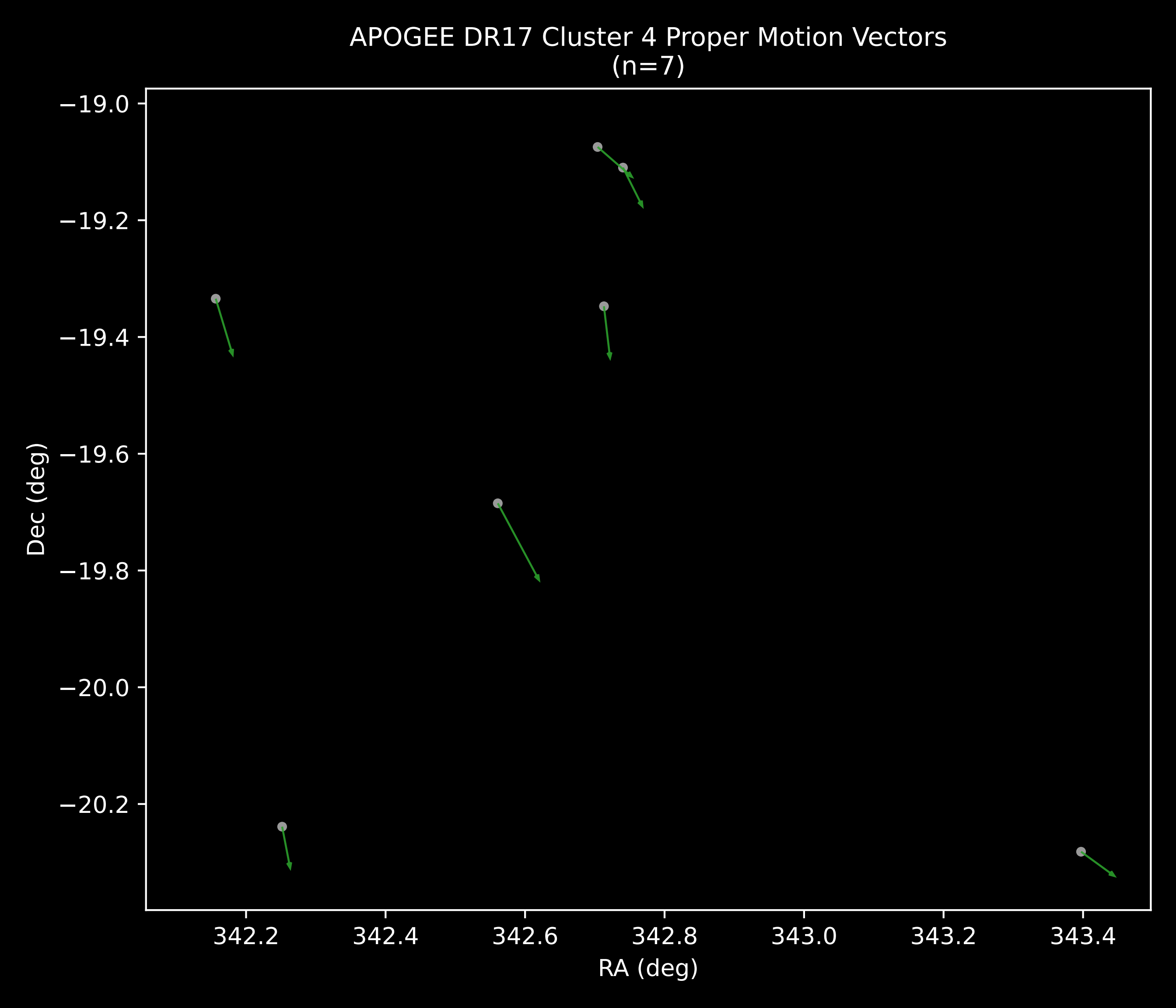
APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=17)
−19.00
−19.25
−19.50
−19.75
−20.00
−20.25
−20.50
−20.75
−21.00
Dec (deg)
342.2
342.4
342.6
342.8
343.0
343.2
RA (deg)



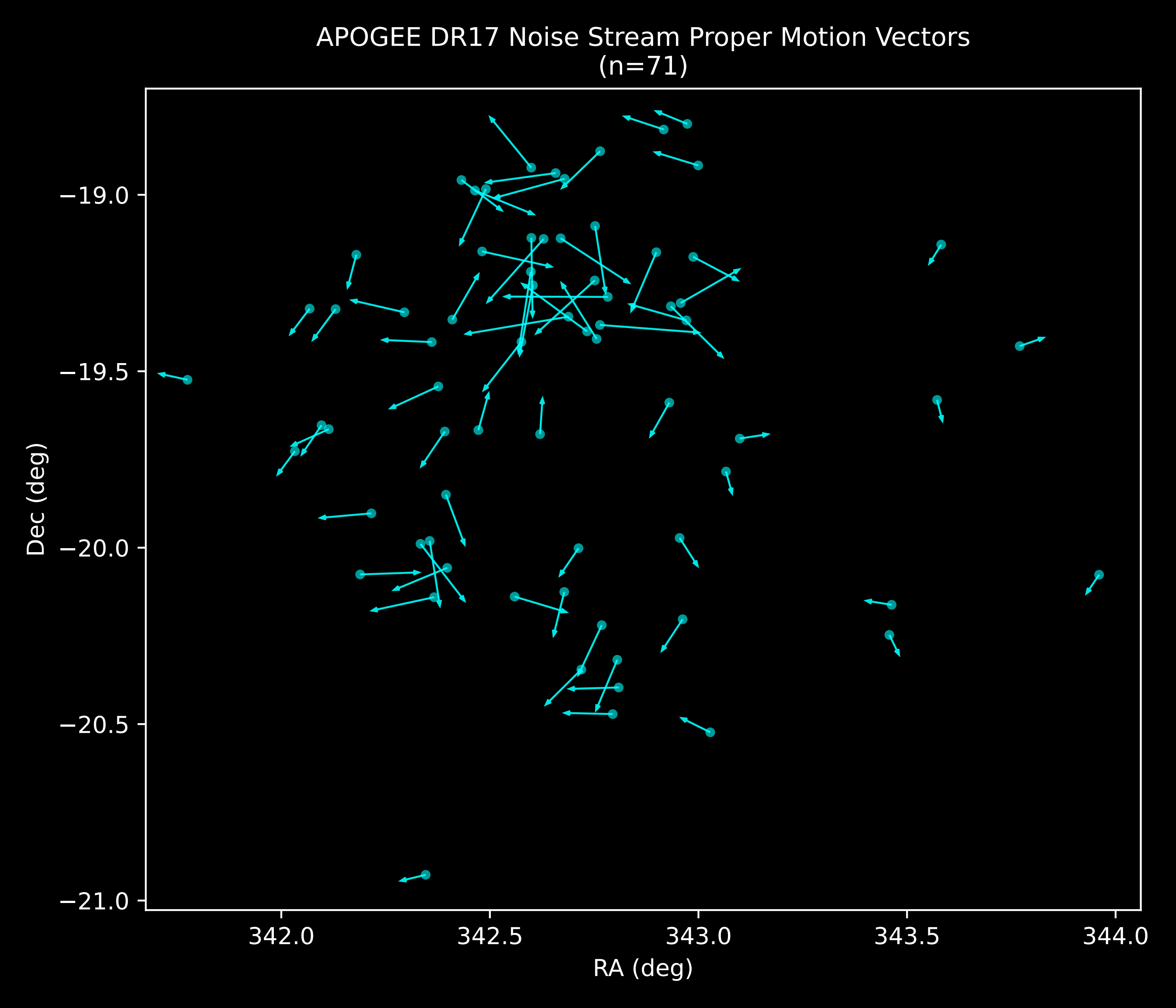
APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=6)
Dec (deg)
−19.1
−19.2
−19.3
−19.4
−19.5
−19.6
−19.7
−19.8
RA (deg)
341.5
342.0
342.5
343.0
343.5


APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=7)
−19.0
−19.2
−19.4
−19.6
−19.8
−20.0
−20.2
−20.4
Dec (deg)
342.0
342.2
342.4
342.6
342.8
343.0
343.2
343.4
RA (deg)


APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=45)
Dec (deg)
RA (deg)
−19.0
−19.2
−19.4
−19.6
−19.8
−20.0
−20.2
−20.4
−20.6
342.00
342.25
342.50
342.75
343.00
343.25
343.50
343.75



APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=7)
−19.0
−19.2
−19.4
−19.6
−19.8
−20.0
−20.2
Dec (deg)
342.2
342.4
342.6
342.8
343.0
343.2
343.4
RA (deg)



APOGEE DR17 Noise Stream Proper Motion Vectors
(n=71)
−19.0
−19.5
−20.0
−20.5
−21.0
Dec (deg)
342.0
342.5
343.0
343.5
344.0
RA (deg)